

Prognostics for the Sustainability of Industrial Cyber-Physical Systems: From an Artificial Intelligence Perspective

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Abstract—As industrial cyber-physical systems (ICPS) play an increasingly pivotal role in the new industrial paradigm, their sustainability has become the current research focus. Remaining useful life (RUL) prediction, also known as prognostics, is critically significant for the sustainability of ICPS. The prognostics involve utilizing process monitoring devices within ICPS to acquire real-time operational data. Based on the trends observed in the monitored data, the analysis predicts when potential system failures may occur. Accurate prognostics allow real-time monitoring of the system's health status, which enables early warning of possible faults and system reliability. The primary objective of this paper is to provide readers with a timely survey and review that reveals the current research status, development trends, and common challenges in the prognostics domain within ICPS. From the perspective of artificial intelligence (AI), the paper comprehensively reviews predictive approaches based on stochastic process, machine learning, and their hybrid applications. Through a comprehensive comparison of existing approaches, the paper delves into the strengths and weaknesses of these approaches. Furthermore, facing some cutting-edge issues in existing RUL prediction approaches for ICPS, this paper analyzes some pioneering investigations that have achieved great results. Finally, the paper explores the opportunities and challenges of prognostics from the perspective of artificial intelligence, which aims to drive the sustainability of ICPS.

Index Terms—Prognostics, artificial intelligence, remaining useful life, industrial cyber-physical systems.

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I. INTRODUCTION

CYBER-PHYSICAL systems (CPS) represent a forefront in technological advancements, which mark a milestone in the profound integration of computer science with the physical world [1], [2]. CPS are systems composed of computer, communication, artificial intelligence (AI), and control technologies that are closely integrated, which combines computing power and feedback from the physical world [3], [4], [5]. CPS have widespread applications in advanced manufacturing, transportation, health care, and other fields [6], [7]. Industrial cyber-physical systems (ICPS), a specific domain within CPS, are primarily employed in industrial production and manufacturing environments [8]. Based on CPS, ICPS accomplish the data collection, analysis, monitoring, and control of industrial production processes through deep integration of computer networks, and physical process technologies [9], [10], [11], [12], [13].

Driven by rapid advancements in computer, information communication, and sensor technologies, ICPS are experiencing a growing scale, which reaches unprecedented levels of system integration [14], [15], [16]. Simultaneously, as the structure of ICPS becomes increasingly complex, the connections among its components become tighter. When a component in ICPS fails, the normal operational status of the system will be directly affected, which leads to significant issues such as resource wastage, production interruptions, economic losses, and even personnel injuries. Consequently, the sustainability of ICPS has become a focal point of current research. During the operation, the health status of the ICPS should be monitored in real-time. Furthermore, the possible failures should be accurately predicted. In this sense, the users can be assisted in formulating a reasonable maintenance plan, thereby optimizing the maintenance to ensure the sustainability of the whole ICPS.

Based on the above background, the prognostics, in other words, the remaining useful life (RUL) prediction, has significant research value. The RUL refers to the remaining time between the current moment and the time when the performance cannot be satisfied [17]. Consequently, accurately predicting the RUL is a hot topic of widespread concern for the sustainable development of ICPS. Accurate RUL prediction can provide early warning of faults, which optimizes maintenance strategies,

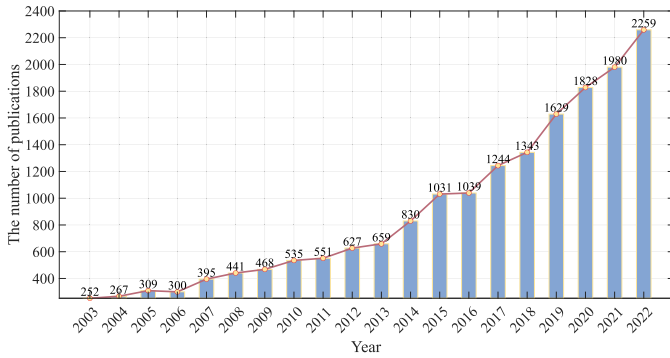


Fig. 1. The number of papers published on the theme of RUL prediction in recent years.

and promotes safety and reliability [18], [19], [20], [21]. On the basis of the statistical data of Web of Science, this article counts the number of papers published on the theme of RUL prediction in recent years, as shown in Fig. 1. From Fig. 1, since 2003, the number of papers published on the topic of RUL prediction has gradually increased.

In recent years, the RUL prediction research has primarily revolved around two directions: damage mechanism modeling and artificial intelligence [22]. The core idea of the damage mechanism modeling approach is to abstract the lifespan damage process of ICPS into a mathematical model. This modeling approach demands substantial prior knowledge of the principles. It is worth mentioning that the applicability of damage mechanism modeling has notable limitations when there are external disturbances, nonlinearity, and uncertainty in ICPS. Consequently, the emergence of artificial intelligence-based approaches for predicting the RUL has become prominent. Dedicated to integrated multi-modal data integration, Chen et al. proposed a cross-domain RUL prediction approach based on adaptive dynamic clustering, which can achieve great prediction result [23]. From the perspective of enhancing model credibility, Li et al. proposed an RUL prediction approach based on the sensor-aware capsnet, which can attribute uncertainty and RUL prediction results to sensor characteristics [24]. Chen et al. also proposed a transfer regression network based on adaptive calibration, which can fully consider the shared features and private features in the target domain. The prediction error in the shared RUL regressor is automatically calibrated to promote the RUL prediction performance [25]. Artificial intelligence-based approaches for RUL prediction include stochastic process, machine learning, and the integration of stochastic process and machine learning [26]. Fig. 2 illustrates the current classification of artificial intelligence-based approaches for RUL prediction.

Recently, some researchers have reviewed RUL prediction approaches from the perspective of artificial intelligence. Table I lists some representative reviews of RUL predictions based on artificial intelligence in recent years, whose research approaches have different focuses. In detail, from the perspective of RUL prediction with the Wiener process, Zhang et al. discussed the current research status of RUL prediction with Wiener process from multi-source variability, nonlinearity, and multivariables,

TABLE I
A REVIEW SUMMARY OF RUL PREDICTION APPROACHES BASED ON ARTIFICIAL INTELLIGENCE

Year	References	Review perspective
2018	Zhang et al. [27]	Stochastic process-based approaches
2020	Dalzocho et al. [28]	Machine learning-based approaches
2021	Wang et al. [29]	Deep learning-based approaches
2022	Ferreira et al. [30]	Machine learning-based approaches
2023	Li et al. [31]	Large-scale foundation models
2023	Chen et al. [32]	Transfer learning-based approaches

which reported its application in optimizing decision-making activities [27]. Dalzocho et al. systematically analyzed the works related to machine learning-based predictive maintenance, under the influence of Industry 4.0 in the past five years, which emphasized the applications and challenges [28]. Taking lithium-ion batteries as the research object, Wang et al. reviewed the RUL prediction approaches based on deep learning, which defined specific criteria to evaluate the modeling accuracy of deep learning [29]. Ferreira et al. discussed the advantages, disadvantages and challenges of RUL prediction approaches based on machine learning. Furthermore, a general RUL prediction framework based on machine learning is also elaborated [30]. In the context of the widespread applications of large models, Li et al. comprehensively reviewed the technical routes and future research directions of PHM with the aid of large models [31]. From the perspective of industrial applications, Chen et al. took rolling bearings as the research object, which conducted an in-depth investigation of the RUL prediction approaches based on transfer learning. In this sense, the suggestions are provided for practical industrial applications [32].

Despite the emergence of artificial intelligence-based approaches for RUL prediction as a burgeoning topic in the academic field, there is currently no comprehensive review from the perspective of artificial intelligence in ICPS. To address this gap, this paper provides a comprehensive review of RUL prediction approaches from the perspective of artificial intelligence in ICPS. This paper can be succinctly summarized by highlighting the following key contributions:

- Facing the sustainability of ICPS, this article analyzes the development trends, research ideas, and recent related works based on stochastic process, machine learning, and their integration approaches for the RUL prediction, from the perspective of artificial intelligence. Furthermore, the advantages and potential limitations of these approaches in existing research are discussed.
- Facing some cutting-edge issues in existing RUL prediction approaches for ICPS, this paper analyzes some existing technologies that have achieved great results, which can provide some incentives for existing RUL prediction research.
- This paper explores the potential opportunities and challenges for prognostics from the perspective of artificial intelligence. The primary objective of this paper is to enhance the research on RUL prediction, which contributes to the sustainability of ICPS.

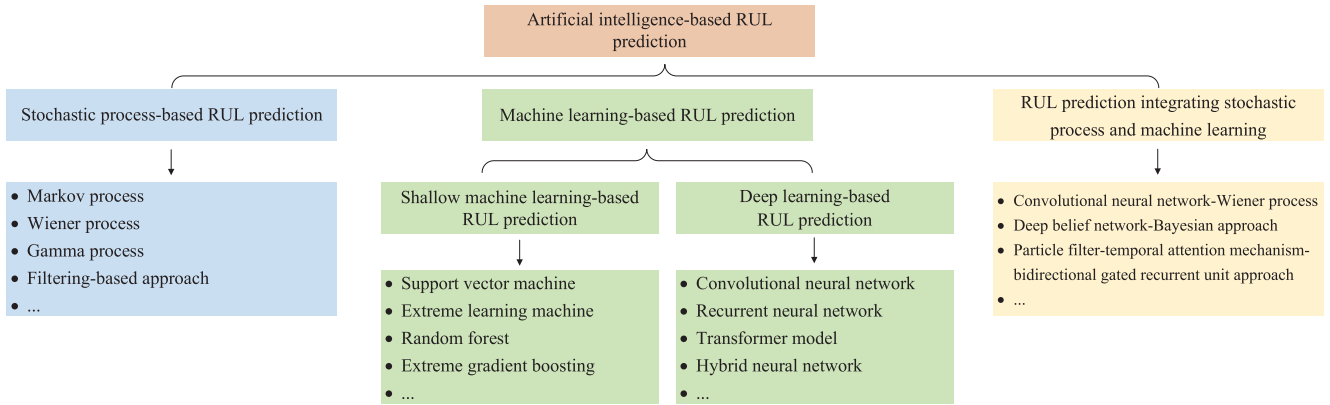


Fig. 2. The current classification of artificial intelligence-based approaches for RUL prediction.

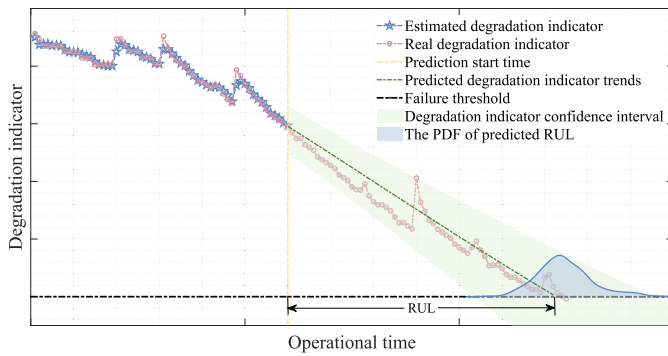


Fig. 3. The general idea of the RUL prediction based on stochastic process.

The reminder of the paper is organized as follows. Section II introduces the RUL prediction approaches based on stochastic process. Section III introduces the RUL prediction approaches based on machine learning. Section IV introduces the RUL prediction approaches integrating stochastic process and machine learning. Section V describes some pioneering investigations in RUL prediction. Section VI discusses the opportunities and challenges. Section VII summarizes the paper.

II. RUL PREDICTION BASED ON STOCHASTIC PROCESS

The RUL prediction based on stochastic process integrates the prior knowledge of system degradation, which constructs a random evolution model of the ICPS. Consequently, the probability density function of failure time is learned. The general idea of the RUL prediction based on stochastic process is shown in Fig. 3. Based on probabilistic stochastic process theory, the evolution rules of degradation indicators are modeled, so that the RUL distribution can be deduced in the form of a probability distribution. On this basis, not only can the point estimation of the RUL be obtained, but the uncertain probability density interval also can be calculated. Currently, common stochastic process approaches include Markov process [20], Wiener process [33],

Gamma process [34], and filtering-based approaches [35]. This section will provide an in-depth explanation, shown as follows:

1) **Markov Process:** As a typical stochastic process, the Markov process has the characteristic of memorylessness [36], [37], [38]. In other words, the future state only depends on the current state, which is not affected by the past state. In the RUL prediction, a Markov chain model is constructed to describe the evolution of the degradation system state. By analyzing the transition probability between states, the future state is predicted, and the RUL distribution is further estimated. To promote the RUL prediction accuracy, Zhu et al. integrated the Grey-Markov chain model with an improved Gaussian process model, which decomposed the system's degradation process using empirical mode decomposition with online calibration [38]. Considering the influence of operational conditions on the degradation process, Tang et al. designed a discrete-time Markov chain to describe the dynamic system degradation process, which quantified the influence of operational conditions through degradation rates to improve the reliability of aerospace systems [20].

2) **Wiener Process:** As a continuous-time stochastic process, the Wiener process is often used to describe random changes in degradation systems, such as noise, and disturbance [39], [40], [41]. A vital feature of the Wiener process is continuity. Furthermore, the stochasticity is represented by standard Brownian motion, which enables the Wiener process to model uncertainties in degenerate systems. Yu et al. proposed a degradation model based on the generalized Wiener process. In addition, a recursive Bayesian filtering approach is designed to update the drift distribution and predict the RUL of cutting tools [41]. Committed to deep feature fusion, Li et al. integrated deep learning approaches and Wiener process, to predict the RUL of milling tools [33]. Considering the dependence between operational conditions and degradation rates in degradation modeling, Guan et al. proposed an RUL prediction approach based on the dependence Wiener process, which realizes the collaborative parameter estimation through the expectation maximum and the Bayesian algorithm [42].

3) **Gamma Process:** The Gamma process is often used to describe the random growth of non-negative continuous random variables [43], [44], [45]. This process consists of random

variables obeying a gamma distribution, which describes the intensity of events in a degradation system at different time points. Moreover, the Gamma process has non-homogeneous characteristics, which gives the process certain advantages in degradation modeling. Dedicated to decreasing the influence of random fluctuations on RUL prediction, Wang et al. proposed an RUL prediction approach based on isotonic regression and the Gamma process, which designed a health stage identification strategy through the gradient within the sliding window [43]. Considering that the practical system may be affected by measurement errors, Liu et al. designed a degradation increment related to time and state, which adopted a modified Gibbs sampler to estimate the implicit degradation state [34].

4) *Filtering-Based Approaches*: On the basis of the observation data of the degradation system, the filtering-based approaches can estimate the implicit state, which considers the noise and uncertainty in the degradation process. These approaches aim to provide the optimal estimation of the degradation system state. Common filtering-based approaches include Kalman filter (KF) [46], particle filter (PF) [47], unscented particle filter (UPF) [48], etc. Li et al. proposed a PF algorithm based on a non-parametric degradation model, which constructed a multi-output Gaussian regression to accomplish the relationship between capacity degradation and battery health indicators, thereby predicting the battery's RUL [47]. Yu et al. modified the conventional autocorrelation kernel regression from the perspective of distance similarity and spatial similarity. On this basis, a monotonic constrained PF is proposed to update the random model coefficients, thereby promoting the RUL prediction accuracy [35].

From the above literature analysis, the stochastic process RUL prediction approaches are based on the theory of probability and statistics, which models the evolution law of performance degradation indicators from the perspective of the stochastic model. Probability distribution has great advantages in quantifying the uncertainty of RUL prediction, which has strong model interpretability. In this sense, RUL prediction based on stochastic process has widespread applications in ICPS. Nevertheless, when predicting the RUL, these approaches need to meet the following two requirements:

- 1) *Prior knowledge*: Similar to the RUL prediction approaches based on damage mechanism modeling, the RUL prediction approaches based on stochastic process demand certain prior knowledge to complete the model construction. Conversely, in practical ICPS, it is challenging to acquire prior knowledge, which limits the application of the RUL prediction approaches with stochastic process.
- 2) *Model assumption*: The RUL prediction approaches based on stochastic process usually obey some specific model assumptions, such as random state transitions, continuous degradation processes, etc. Nevertheless, various uncertain situations may exist in ICPS, such as nonlinearity, multi-modal degradation, strong external interference, etc. Consequently, some specific assumptions may not be fully satisfied. The RUL prediction accuracy with

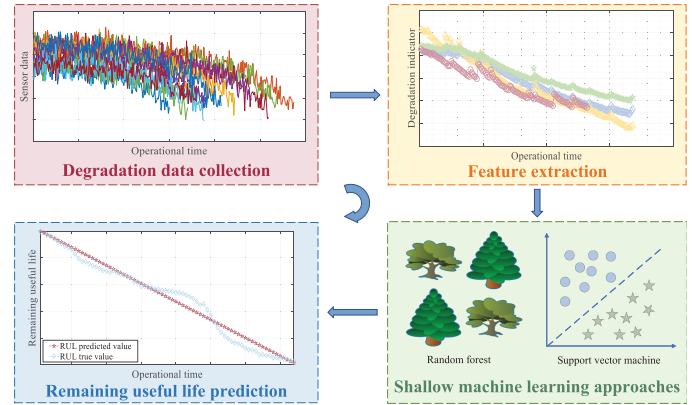


Fig. 4. The basic framework of the RUL prediction based on shallow machine learning.

stochastic process may be decreased when ICPS violate the model assumptions.

III. RUL PREDICTION BASED ON MACHINE LEARNING

The RUL prediction based on machine learning constructs a learning algorithm model, which constructs the mapping relationship between sensor data and the RUL to accomplish direct end-to-end prediction [49]. Alternatively, the machine learning-based approaches can also learn the evolution rules of degradation features, which extrapolate the degraded features to the failure threshold to obtain the RUL, according to the trend extrapolation strategy. To sum up, without any prior knowledge, the RUL prediction based on machine learning can automatically learn feature representations from the degradation data of ICPS, which has great expressiveness and adaptability. According to the depth of the learning model, machine learning-based approaches can be further divided into shallow machine learning approaches and deep learning approaches. The shallow machine learning approaches usually adopt conventional feature extraction methods, while deep learning approaches automatically learn feature representations through multi-layer neural networks. The deep learning approaches generally perform better on large-scale degraded datasets, which require more computing resources.

A. Shallow machine learning approaches

The basic framework of the RUL prediction based on shallow machine learning can be shown in Fig. 4. Firstly, the historical degradation data are obtained through the sensor monitoring equipment in ICPS. The features related to health status are extracted. Furthermore, these features are used to train a shallow machine learning model. In this sense, a mapping relationship between degradation features and RUL can be constructed to complete the offline model training. Finally, the online RUL prediction can be achieved based on online sensor data.

Currently, typical RUL prediction with shallow machine learning approaches include support vector machine (SVM) [50], extreme learning machine (ELM) [51], random forest

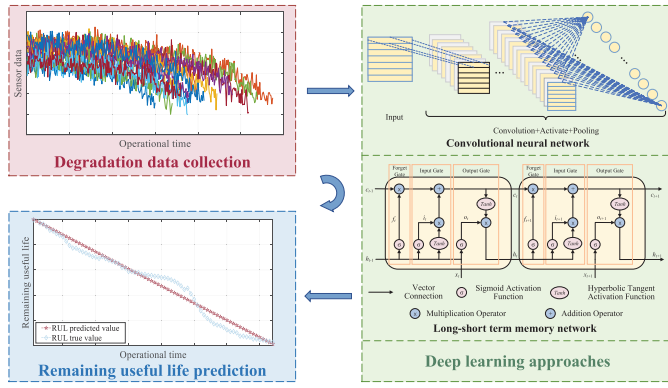


Fig. 5. The basic framework of the RUL prediction based on deep learning approaches.

(RF) [52], extreme gradient boosting, [53], etc. To eliminate the individual differences and to promote sensitivity to early defects, Yan et al. proposed an RUL prediction approach through SVM to evaluate the degradation stage. Furthermore, a hybrid degradation tracking model is designed to further complete the RUL prediction [50]. Committed to multivariable regression and few-shot prediction, Pan et al. proposed a two-stage prediction approach based on ELM, which designed a feedback model for short-term RUL prediction [51]. In view of the advantages of the RF at the feature selection level, Yang et al. adopted partial charging data and discharging data under limited degradation history conditions to predict the RUL of lithium-iron phosphate batteries [52].

B. Deep learning approaches

Although shallow machine learning approaches have achieved certain success in RUL prediction, they still have limitations. Specifically, shallow machine learning approaches rely on manual design in the feature extraction process, which may cause subjectivity in feature selection. In addition, shallow machine learning approaches may face challenges in model complexity, and computational efficiency, when processing large-scale data in ICPS. To overcome these limitations, the RUL prediction based on deep learning approaches has gradually emerged in recent years, which can automatically learn feature representations, and have more robust expressive capabilities and adaptability. Specifically, convolutional neural network (CNN) [54], recurrent neural network (RNN) [55], Transformer model [56], and hybrid neural network [57], have expanded the scope of artificial intelligence RUL prediction approaches. The basic framework of the RUL prediction based on deep learning approaches is shown in Fig. 5, which shows the deep learning approaches, CNN, and LSTM as examples.

CNN is composed of several convolutional layers and pooling layers, which has some advantages, such as sparse representation, weight sharing, and pooling operations. Consequently, CNN has widespread applications in image processing, natural language processing, and computer vision. Regarding ICPS,

many types of CNNs have been used for RUL prediction, including multi-scale CNN [58], graph CNN [59], and multi-channel CNN [60], etc. Recently, Ruan et al. proposed a weight ranking network, which designed a lightweight CNN through model pruning to predict the RUL of bearings [61]. To promote the efficiency of feature extraction, Zhang et al. proposed a CNN with a temporal attention mechanism for RUL prediction, which considers the contribution of different channels of input features [62]. It is worth mentioning that CNN does not have a structure that is specifically designed to capture long-term dependencies in time series data. Consequently, it may not capture the correlation between time sequences well.

RNN is a type of network that is specially designed to process serialized data, the essence of which is to introduce a cyclic structure so that the network can remember historical information [63]. Furthermore, the long-short term memory network (LSTM), and gated recurrent unit (GRU) aim to solve the problem that conventional RNN is difficult to capture long-term dependencies [63]. Compared with LSTM, GRU decreases the structure by one gate structure, which reduces computing resources. In addition, GRU incorporates update gates and reset gates to control the flow and forgetting of information. By introducing gate units, the information flow is effectively controlled, thereby better capturing long-term dependencies. Qin et al. decomposed the data to obtain slow-variation features. Moreover, the LSTM was introduced into the capsule network, which captured the correlation of time for the RUL prediction [64]. Hu et al. integrated the deep bidirectional RNN with ensemble learning, which iteratively trained multiple regression decision trees, to adjust the weight of the base learning model for the RUL prediction [65].

As a network architecture based on the multi-head attention mechanism, the essence of Transformer is to use self-attention to process sequence data [66]. At the same time, Transformer can process large-scale data more efficiently through parallel computing, which helps to deal with large amounts of time series data in RUL prediction for ICPS. Ren et al. proposed a dynamic time length Transformer for the RUL prediction that can adaptively learn sequence representations of different lengths, which calculated the confidence level of the results by designing a confidence strategy [66].

Hybrid neural networks can make up for the shortcomings of existing single types of deep networks by combining multiple types of neural networks. Currently, existing hybrid neural network integration approaches include CNN-LSTM [67], CNN-RNN [55], CNN-GRU [68], etc. Committed to the integration of temporal information and spatial information, Zraibi et al. combined CNN and LSTM networks for the RUL prediction of lithium-ion batteries [67]. In addition, Wilberforce et al. integrated CNN and bidirectional RNN for the RUL prediction of proton exchange membrane fuel cells [55]. Considering that the serial connection of the spatial information extraction sub-network and the temporal information extraction sub-network of the hybrid neural network may affect each other, our team proposed a parallel deep neural network that combines a one-dimensional CNN and a bidirectional GRU from a parallel

perspective. The aircraft turbofan engine and milling prop wear datasets were used for the approach verification [68].

The RUL prediction, with the aid of deep learning approaches, can automatically learn feature representations from degradation data, which have stronger adaptability for the ICPS. It should be noted that the RUL prediction based on deep learning approaches faces challenges in the following directions:

- 1) *Degradation data quality constraints*: Most devices must work in a healthy state. In this sense, obtaining many high-quality degradation data samples is a great challenge. Nevertheless, deep learning approaches have certain limitations, when degraded data quality is limited. The reason is that the learning of degradation data by deep network relies on large-scale labeled samples. In this sense, the limited conditions for degradation data quality, such as few-shot samples, and variable operational conditions, may prevent deep neural networks from establishing accurate mapping relationships.
- 2) *Deep neural network interpretability*: Deep neural networks are essentially black box models. In other words, the decision-making process and internal model mechanism are difficult to explain. The reason is that deep neural networks include a large number of neurons, which makes the decision-making process very complex to explain through simple mathematical equations. Consequently, how to promote the interpretability of the neural network is a great challenge.
- 3) *Degradation data privacy protection*: Due to commercial privacy and sensitivity, the industrial equipment manufacturers or operators in ICPS restrict the sharing of equipment performance data, which results in limited access to degradation information. Nevertheless, deep learning demands a large amount of data for training, which may involve sensitive device information. Consequently, how to predict the RUL under privacy protection of degradation data under multi-client collaborative training requires further research.

IV. RUL PREDICTION INTEGRATING STOCHASTIC PROCESS AND MACHINE LEARNING

Through the above discussion, machine learning-based approaches focus on learning feature representations from degradation data, which constructs a nonlinear mapping relationship between features and the RUL. On the other hand, stochastic process-based approaches have great interpretability, which can deduce the RUL distribution with a probability distribution to describe the uncertainty. Consequently, by integrating stochastic process and machine learning, it is possible to fully utilize the powerful nonlinear modeling capabilities of machine learning while describing the degradation process of ICPS [69]. The overall idea of the RUL prediction approach that integrates stochastic process and machine learning is shown in Fig. 6. Through the data collected by sensors, the machine learning approaches are first adopted to extract the deep degradation features of ICPS. Furthermore, appropriate features are selected for stochastic process modeling through feature engineering.

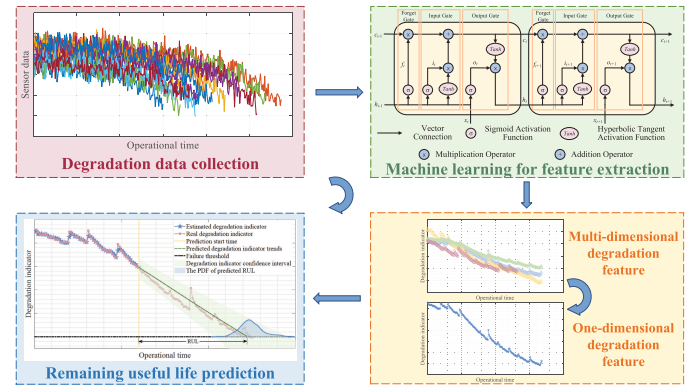


Fig. 6. The overall idea of the RUL prediction approaches integrating stochastic process and machine learning.

Finally, the system degradation process is modeled through the stochastic process to predict the RUL.

At present, some researchers have carried out research on the RUL prediction approaches that integrate stochastic process and machine learning. Zhang et al. integrated CNN and stochastic process approaches, which established a nonlinear multi-level Wiener process to describe the system degradation process [70]. Committed to the uncertain RUL description, Pei et al. designed a continuous deep belief network to mine the implicit information in the degradation data, which adopted a diffusion process to construct an evolution system model. In this process, the Bayesian approach is combined to achieve online parameter estimation of the diffusion process [71].

Although there are research advantages in RUL prediction approaches that integrate stochastic process and machine learning, these approaches do not essentially solve the challenges faced by machine learning, such as degradation data quality constraints, deep neural network interpretability, and degradation data privacy protection, etc. Furthermore, for the integration approach of the stochastic process and machine learning, whether the degradation features extracted by machine learning can adapt to the stochastic process model still requires in-depth discussion. Consequently, the integration strategy of stochastic process and machine learning needs further research to ensure that the advantages of each approach. The methods, such as dynamic weight adjustment, and adaptive fusion can be considered to adapt to different scenarios of ICPS.

V. SOME PIONEERING INVESTIGATIONS IN RUL PREDICTION

To better expand the discussion of the pioneering investigations in RUL prediction, this section will describe separately according to the approach classification in Section I.

A. The Pioneering Investigations in RUL Prediction Based on Stochastic Process

In the RUL prediction task, the system degradation process may be affected by various factors, for instance, material quality, environmental condition, and usage strategy, which results in parameter uncertainty of degradation model [72]. Consequently,

it is necessary to accurately estimate these degradation model parameters to avoid a decrease in RUL prediction performance. Moreover, some existing stochastic process-based approaches are subject to the strong Markov property. The strong Markov property means that the future state of a stochastic process only depends on its current state, which does not depend on its past state [73]. In this sense, when the strong Markov property is satisfied, the evolution process is memoryless, so that the information on past state may not affect future states. Based on these observations, regarding the RUL prediction with stochastic process, how to adaptively update the degradation model parameters through the online sensor data obtained in ICPS, and to overcome the strong Markov property, require in-depth research.

Driven by the above background, our team has conducted in-depth research. Without the need for historical degradation data, our team proposed a dynamic degradation model based on the Wiener process to describe the systematic degradation process. Furthermore, regarding the parameter update in stochastic degradation models, we proposed an expectation maximum for degradation model parameter estimation, which can describe the uncertainty of model parameters to promote the RUL prediction performance. Facing the strong Markov constraints in conventional stochastic process approaches, we designed a backward smoothing method based on the Rauch-Tung-Striebel theory, which rationally utilizes the system degradation data from the initial state to the current moment [46]. The above work starts from the perspective of the linear Wiener process describing the system degradation mode. It is worth mentioning that ICPS inevitably has nonlinear degradation characteristics in the actual operational process [74]. Furthermore, in engineering practice, the evolution of health indicators may occur outliers, due to sensor failure (sensor misalignment or deviation), external interference (electromagnetic interference or vibration) [75]. Consequently, on the basis of [46], our team proposed an RUL prediction approach based on a nonlinear degradation model. In detail, focusing on the uncertain parameters of dynamic degradation models, the expectation maximum with UPF is designed for adaptive degradation model parameter estimation. In addition, the proposed approach can identify the abnormal points in the system degradation process with the aid of statistical learning, which facilitates timely correction of the degradation model to promote the RUL prediction performance [48].

B. The Pioneering Investigations in RUL Prediction Based on Machine Learning

Recently, machine learning-based approaches have adopted collected historical data for analysis, which seek implicit system information by constructing a learning algorithm model [76]. The machine learning-based RUL prediction approaches have two interesting directions:

1) **Neural Network Sensitivity:** As the complexity of deep network models increases, the number of parameters, and the amount of information are improved, which promote the ability

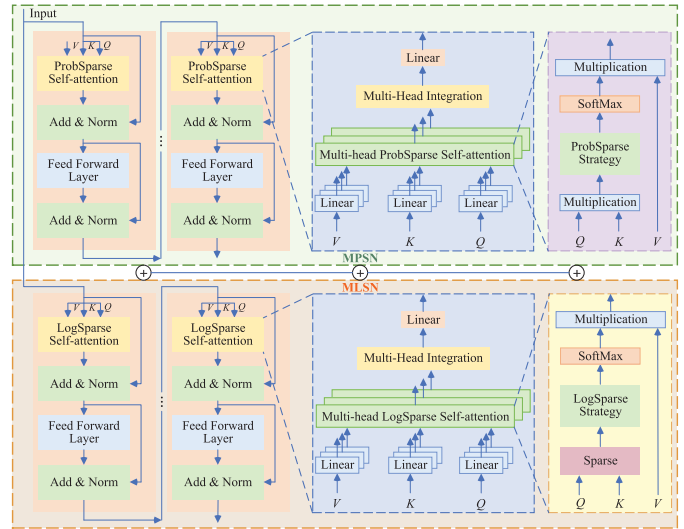


Fig. 7. The framework of the dual sparse self-attention network.

of the model to approximate nonlinear relationships. Nevertheless, it also brings about the problem of information overload, resulting in a waste of computing resources. How to address the challenge of information overload, concentrating limited computing resources on crucial information requires further investigation [77]. To tackle the challenge of focusing on critical regions, without prior knowledge of the target system, our team explored a key region-focus approach based on self-attention mechanisms. A network layer based on a temporal self-attention mechanism is designed, which assigns higher weights to significant time steps, and lower weights to less significant ones. In this sense, the sensitivity to critical areas within the neural network input time window is promoted [78].

On the other hand, considering existing RUL prediction approaches, for instance, Transformer, adopts a global self-attention mechanism to focus on crucial regions of input data, which has limitations in lightweight modeling. Specifically, conventional Transformer with dot-product self-attention may reduce sensitivity to significant areas due to the global query relationships [79]. Consequently, our team focused on the balance between computational efficiency and model lightweight design, which investigates the construction of lightweight attention mechanism networks. From the perspectives of probability sparsity and logarithmic sparsity, we have designed a multi-head ProbSparse self-attention network (MPSN), and a multi-head LogSparse self-attention network (MLSN). By integrating the results of these dual sparse self-attention networks, the RUL prediction of ICPS can be achieved [80]. The framework of the dual sparse self-attention network can be shown in Fig. 7.

2) **Cross-Domain RUL Prediction:** Conventional supervised machine learning approaches for the RUL prediction assume the necessity of consistent distribution between offline training data and online data, i.e., from the same data domain, which is challenging to meet in real-world tasks [81]. In practical ICPS operations, the variations in equipment configuration (component additions or removals, load adjustments), and operational

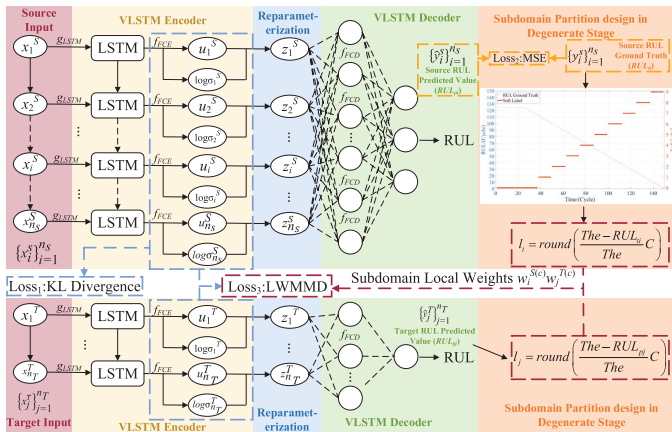


Fig. 8. The framework of the fine-grained subdomain transfer network.

changes (product modifications, process alterations) may make it challenging to maintain a constant operational condition [82]. Furthermore, different devices may exhibit distinct failure degradation patterns, leading to varied distributions in degradation data. It is worth mentioning that users possess degradation data and label information from only one data domain, which lacks the degradation information for ICPS after domain shifts. Consequently, further research is needed on how to achieve RUL prediction under cross-domain conditions [83], [84], [85]. Transfer learning approaches have broad applicability in addressing cross-domain scenarios [86].

Existing domain adaptation approaches focus on the global alignment of the source and target domains. Nevertheless, in the degradation process of ICPS, different degradation stages exhibit distinct distributions in data representation. Relying on global domain adaptation might lead to a loss of distinction in fine-grained features among sub-domains representing distinct stages, thereby decreasing the effectiveness of RUL prediction. Based on these observations, our team proposed an RUL prediction approach based on a fine-grained subdomain transfer network [87]. We considered the impact of variations in the feature distribution of different degradation stages, which captures fine-grained features between sub-domains representing different stages. In this sense, the differences in the distribution of source domain data and unlabeled target domain data are evaluated, which promotes the RUL prediction performance. The framework of the fine-grained subdomain transfer network can be shown in Fig. 8.

On the other hand, most existing artificial intelligence-based approaches for RUL prediction are tailored for centralized ICPS. Nevertheless, in practical engineering, there exists a category of distributed ICPS, which across vast spatial areas, with workplaces often situated at different locations. The spatial dispersion of distributed ICPS adds complexity to resource sharing, and collaborative work. Some researchers have adopted distributed federated learning-based RUL prediction approaches without cross-domain [88], [89], [90]. Conversely, the impact of cross-domain scenarios requires in-depth research. Focusing on the privacy requirements of distributed federated multi-client

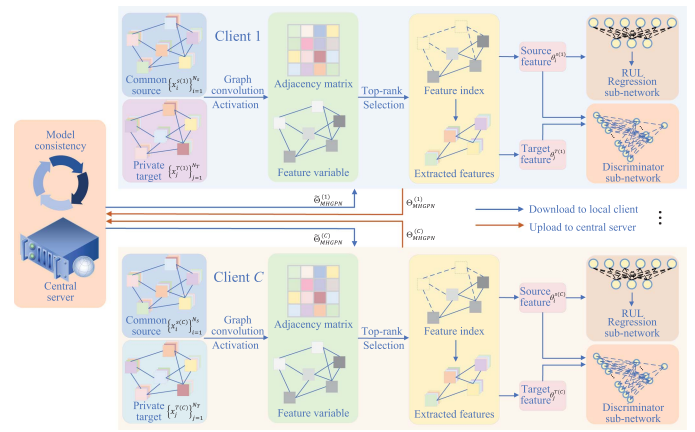


Fig. 9. The framework of the distributed graph adversarial transfer network.

collaborative training, our team proposed an RUL prediction approach based on a distributed graph adversarial transfer network, considering the cross-domain influence on degradation data held by each federated client with adversarial learning [91]. In addition, this approach ensures inter-federate information exchange under data privacy conditions, which designs network optimization strategies in the distributed federated training paradigm. Consequently, the RUL prediction for the target data domain in a distributed federated paradigm can be accomplished. The framework of the distributed graph adversarial transfer network can be shown in Fig. 9.

C. The Pioneering Investigations in RUL Prediction Integrating Stochastic Process and Machine Learning

A key aspect of integrating stochastic process and machine learning approaches is how to leverage the advantages, to enhance the predictive RUL performance of ICPS. Recently, our team addressed this crucial point by proposing an RUL prediction approach for lithium-ion batteries integrating stochastic process and machine learning [92]. Specifically, an attention mechanism network from reference [78] is utilized, which integrates the battery capacity into time windows for future trend prediction. Subsequently, a stochastic process model based on the PF algorithm is constructed to describe the degradation process of lithium-ion batteries. Regarding online RUL prediction, the predicted health indicators from the trained attention mechanism network are used as the posterior values for the PF model at the current moment. This strategy guides updating PF model parameters, facilitating the interaction between these two approaches. Moreover, the ensemble of particles is used to describe the RUL uncertainty. The framework for integrating the stochastic process and machine learning approach is shown in Fig. 10.

VI. OPPORTUNITIES AND CHALLENGES

The examination of current research and the analysis of RUL prediction indicate the increasing prominence of artificial intelligence-based investigations in this field. In the previous

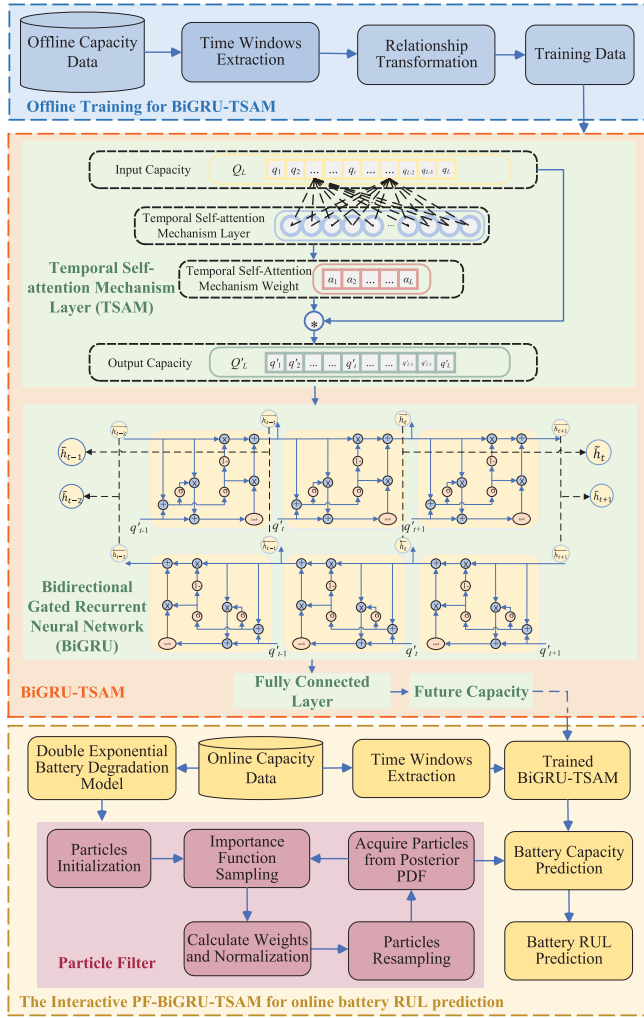


Fig. 10. The framework for integrating the stochastic process and machine learning approach.

sections, we have reviewed recent approaches based on stochastic process, machine learning, and their hybrid applications. In addition to the discussion in the previous sections, this section will further discuss opportunities and challenges for the prognostics for the sustainability of ICPS from the perspective of artificial intelligence. Specifically, some special applications in RUL prediction are discussed, such as (1) Prognostics under degradation data quality constraints, (2) Prognostics based on interpretable deep networks, (3) Prognostics under degradation data privacy, (4) Prognostics based on lightweight model. (5) Prognostics in practical ICPS.

1) Prognostics Under Degradation Data Quality Constraints:

Deep network models rely on large-scale labeled samples to learn degradation data, which have certain limitations, when facing degradation data quality constraints. For instance, due to variable operational conditions and differences in multi-source signals, it has notable limitations in applying historical degradation data to the RUL prediction under new conditions. At present, some researchers have proposed transfer learning approaches to solve this problem [93]. These approaches aim to effectively

transfer the knowledge learned by the model on the source task to another target task, thereby promoting the model's performance on the target task. Specifically, the transfer learning approaches can adopt domain adaptation techniques, including regularized transfer approaches [94], and adversarial transfer learning [95], so that the model can learn the domain differences between the source data and the target data. In addition to transfer learning approaches, deep generative models are also an approach to solve prognostics problems under conditions of degradation data quality constraints. Deep generative models, such as variational autoencoders [96], generative adversarial networks [97], learn data distribution from the latent space. In addition, new synthetic data are generated by sampling the latent space to reduce the impact of data quality constraints. Future research on the RUL prediction approaches for ICPS will focus on the model adaptability in complex environments, and the model transferability in different industrial scenarios.

2) Prognostics Based on Interpretable Deep Networks:

Deep neural networks usually contain a large number of neurons, which lead the decision-making process of the model very complex [78]. Consequently, an interpretable RUL prediction model requires in-depth research, which includes embedding domain knowledge into the predictive model, and designing degradation feature representations with clear meaning. Furthermore, the advanced visualization techniques also await further research [98]. Specifically, exploring the significance of input degradation features, and visual analysis of activations within the model can help explain the model's predictive decision-making process. Finally, the formulation of interpretability evaluation indicators also requires in-depth work, to measure the model performance in interpretable prediction. In this sense, the metrics can be provided for different scenarios to promote the model's interpretability and credibility.

3) Prognostics Under Degradation Data Privacy:

In the ICPS, the data from many devices are dispersed in different locations. Owing to concerns regarding data privacy and security, it is not feasible to store this data centrally. This poses various challenges for conventional centralized approaches for RUL prediction, including data sharing constraints, data security risks, and computing resources limitations. Consequently, distributed federated learning provides an effective solution to the problem of the RUL prediction under dispersed data conditions. The paradigm of distributed federated learning under degradation data privacy is shown in Fig. 11 [91]. In the future, differential privacy technology can be promoted by applying RUL prediction, to ensure effective protection of individual privacy. In addition, federated learning approaches demand further research so that owners of different degradation data in ICPS can jointly construct models in collaborative joint training to promote the model generalization capability.

4) Prognostics Based on Lightweight Model:

With the rise of edge computing, the real-time deployment of lightweight model on edge devices has become significant. In many ICPS scenarios, the computing power, and storage resources of the device are limited. Consequently, the research on lightweight model helps the RUL prediction in resource-constrained environments, which avoids excessive computational burden, and meets the

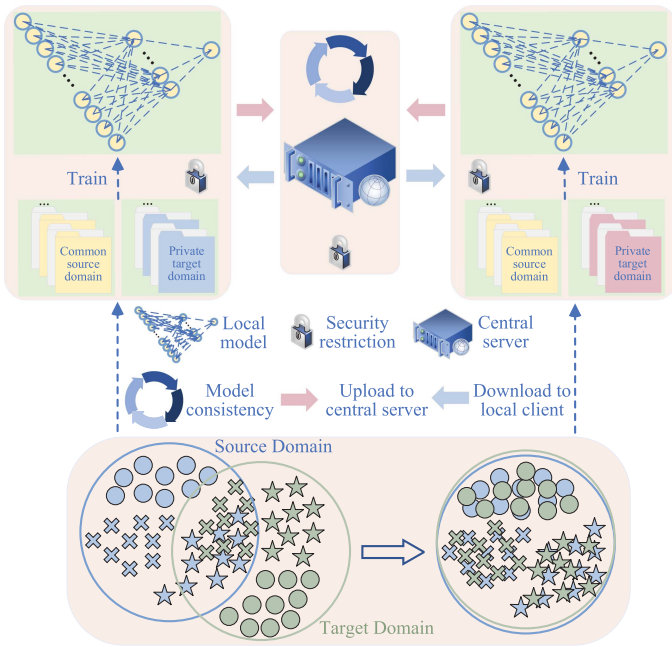


Fig. 11. The paradigm based on distributed federated learning for prognostics in ICPS.

needs of ICPS in real-time. Existing work on lightweight model includes the approaches, such as network pruning [88], sparsification strategy [80], and knowledge distillation [99]. The key to RUL prediction based on lightweight models is the balance between model complexity and performance, which may involve further model optimization structure, and parameter configuration. In the future, the RUL prediction approaches based on lightweight model for ICPS will focus more on lightweight compression of deep networks, intelligent model design on edge and mobile terminals, etc.

5) Prognostics in Practical ICPS: Currently, the degradation datasets used for RUL prediction are usually collected in the laboratory. In practical industrial scenarios, due to the limitations that do not allow ICPS to operate under faulty conditions for extended periods, there are few case studies demonstrating the application of RUL prediction in practical ICPS. In this sense, this paper gives some suggestions for RUL prediction for practical ICPS: a) Data quality and completeness: Ensure the data used are complete, including historical operational data, sensor data, and maintenance records, etc. Inadequate data quality may decrease the model accuracy and reliability. b) Feature selection and extraction: The users need to select the features relevant to the RUL prediction task. The feature selection and extraction should be based on domain knowledge and practical requirements. c) Real-time capability and deployability: Ensure the model can respond promptly and adapt to the operational requirements of the practical ICPS. During deployment, the users need to consider constraints, such as computational resources, data interfaces, and network connectivity of the ICPS. d) Model monitoring and updating: The users need to regularly monitor and evaluate the performance of the model. Since the system operational conditions and environmental factors may change,

the model performance may also vary, which requires timely optimization.

VII. CONCLUSION

Predicting RUL, also known as prognostics, plays a critical role in ensuring the long-term viability of ICPS. By analyzing data collected from monitoring equipment, prognostics can anticipate potential system failures before they occur. This enables real-time health monitoring, allowing for proactive maintenance interventions and ultimately ensuring the safety and reliability of ICPS. This paper delves into the domain of prognostics for ICPS sustainability, specifically focusing on the transformative potential of artificial intelligence. It explores the latest research ideas, development trends, and recent advancements in prognostics, encompassing approaches based on stochastic processes, machine learning, and their powerful hybrid combinations. The paper further delves into a comparative analysis, critically examining the strengths and limitations of these existing approaches to guide future research endeavors. Moving beyond a mere review, the paper delves deeper by analyzing pioneering technologies that have successfully addressed critical challenges in existing RUL prediction approaches for ICPS. These cutting-edge advancements offer valuable insights for further development. Looking towards the future, the paper explores the exciting opportunities and challenges associated with AI-powered prognostics for ICPS sustainability. It delves into specific areas of ongoing research, including the development of robust prognostics methodologies that can function effectively even under constraints of data quality degradation, data privacy concerns, and limitations in computational resources. The paper emphasizes the potential of lightweight models, interpretable deep networks, and novel approaches to address these challenges. This paper aims to stimulate further research and development in this crucial field by comprehensively exploring the current landscape and future prospects of AI-powered prognostics. Its goal is to ultimately contribute to the advancement of sustainable ICPS operation and ensure the continued success of this ever-evolving industrial landscape.

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